Practical 8

Aim: Develop a IoT application which will record the movement and orientation of your phone and give the data back to the PC

Theory:

Home gateway: Home gateway has four Ethernet ports and wireless access point configured with the "Home Gateway" service set identifier (SSID).

All the devices are connected to gateway using config menu in each device, the name of the gateway has to entered.

This can be repeated to all sets of devices that has to be controlled in a home gateway. Ip config menu displays the list of devices connected to the home gateway

Motion Sensor: Motion sensors have a lot of impact in security situations, since they can detect the presence of intruders or sudden changes in the environment.

Motion sensors are small, connected devices that are used to detect and measure movement within a specific area. These sensors are very popular and an essential component of the IoT.

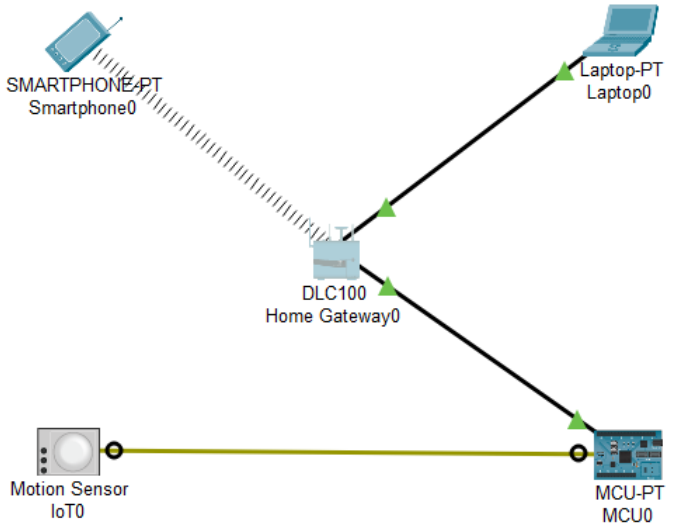
Motion sensors can be used in a variety of applications:

- Home automation.
- Security.
- Mobility

Requirement:

- Smartphone
- Laptop
- Motion Sensor
- MicroController
- Gateway

Step 1: Configure the system

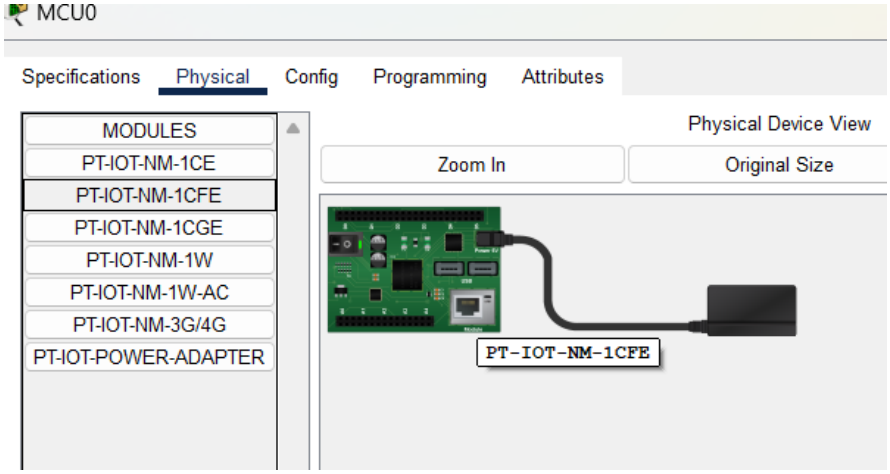


Step 2: Configure the motion Sensor:
Connect the Motion Sensor to the MCU-PT via the IOT Custom Cable.

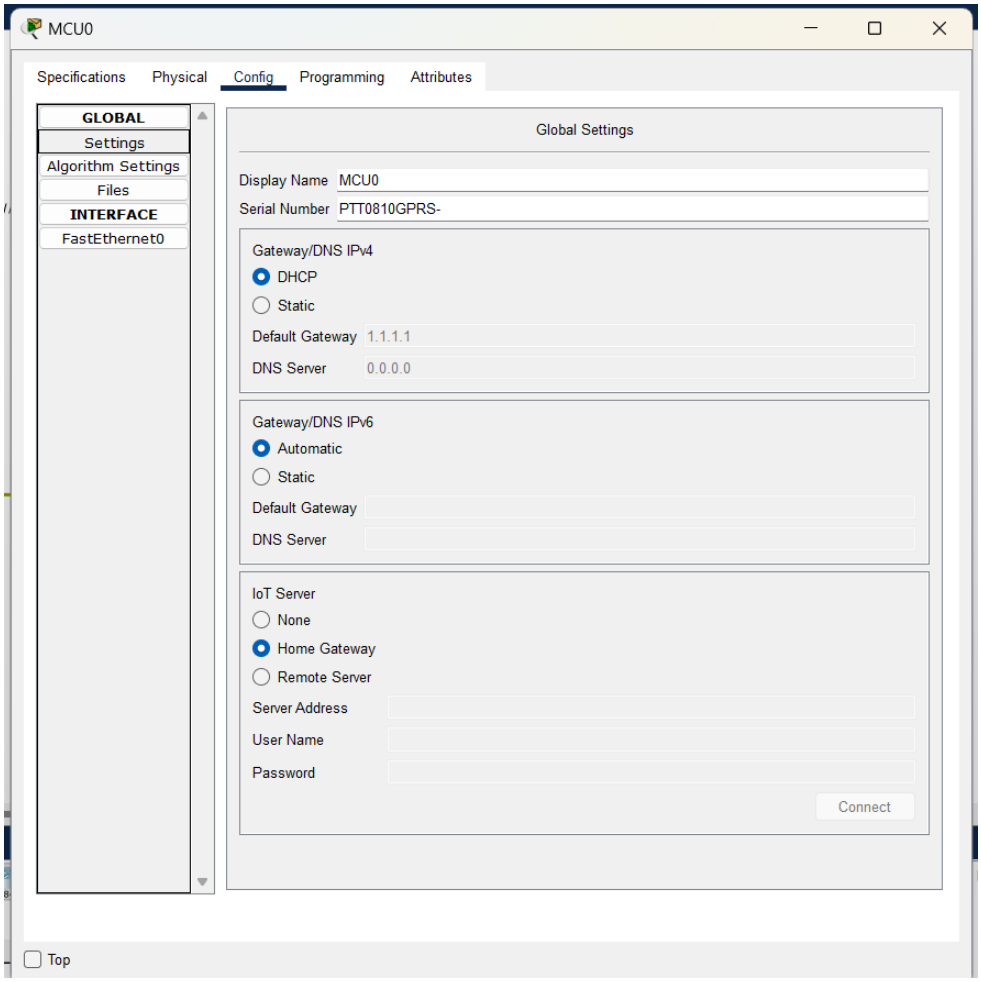
To use the motion sensor press ‘alt’ followed by the hovering of your cursor over the sensor.

Step 3: Configure the Micro Controller:

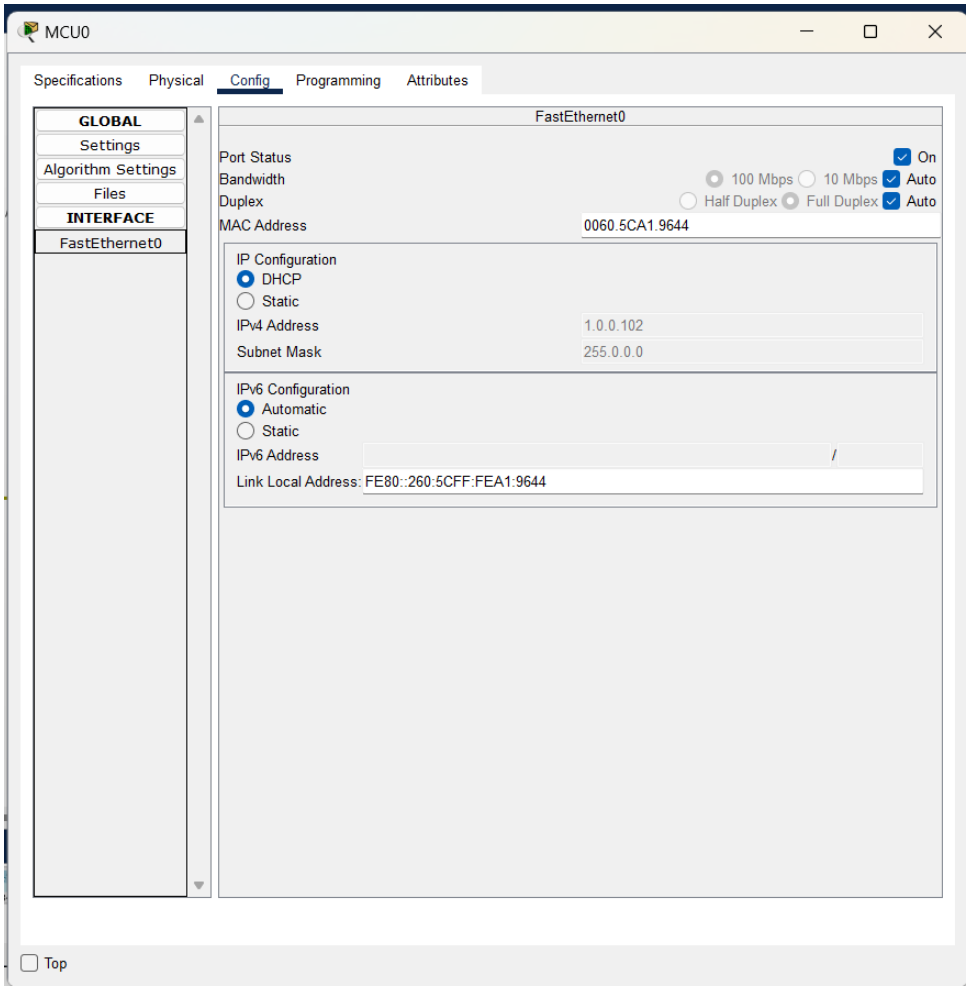
1. Add a CFE port to micro controller.



2. Connect the MCU-PT to the Home Gateway using a copper wire and configure its settings as below.



3. Fast Ethernet setting in MCU:



4. Now we will write the code for the micro controller

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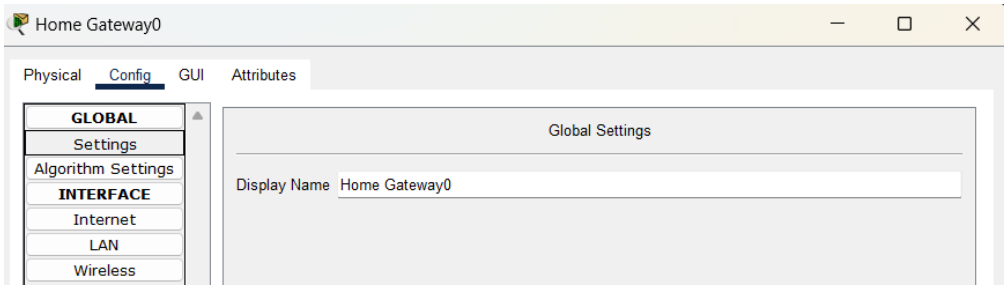
```
function setup()
{
    pinMode(0, INPUT);

    IoEClient.setup({
        type: "MicroContoller",
        states: [{
            name: "Motion Sensor",
            type: "string"
        }]
    });
}

function loop(){
    var motion = digitalRead(0);

    Serial.println(motion);

    if(motion == 1023){
        IoEClient.reportStates(["Movement Detected"]);
    }
    else {
        IoEClient.reportStates(["No Movement Detected"]);
    }
}
```

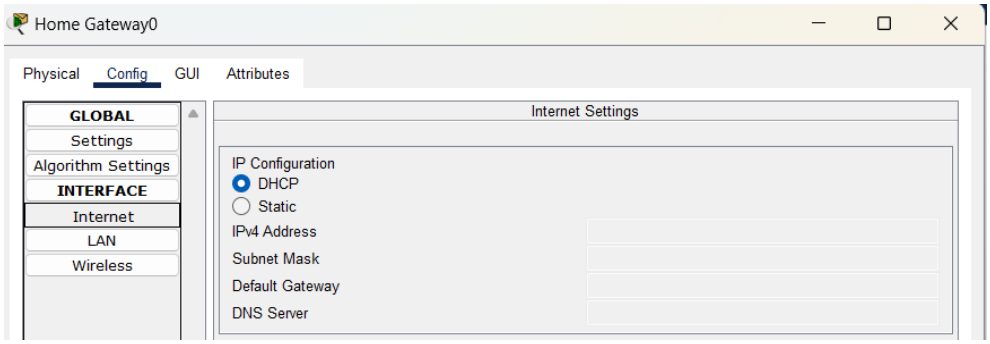
Configuration for the micro controller is done.

Step 4: Configure the Home Gateway:

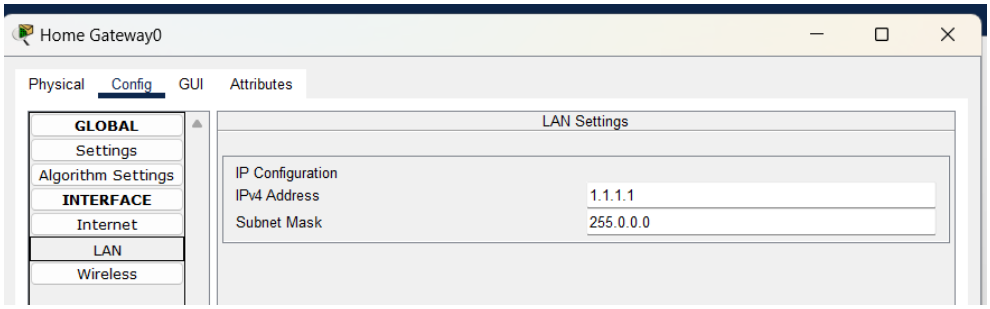
1. Display Name:



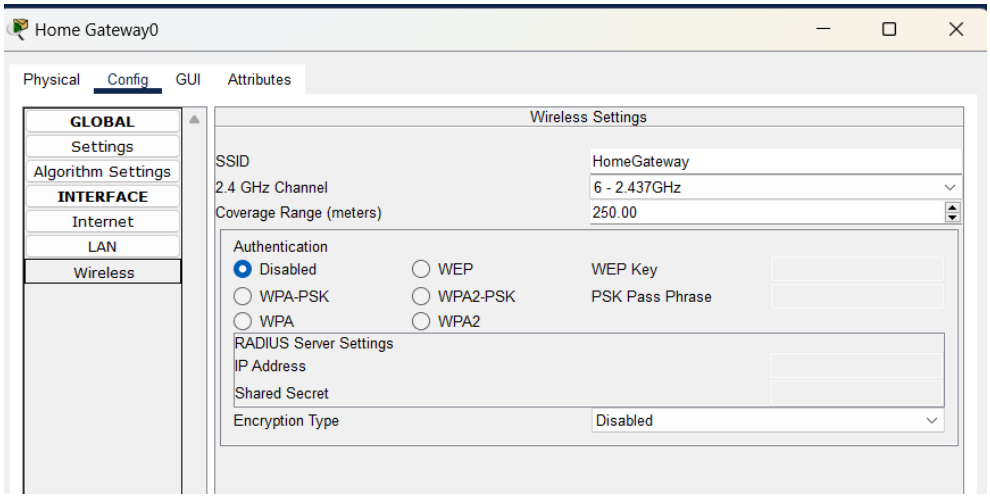
2. Internet setting in Home Gateway:



3. LAN setting in Home Gateway:

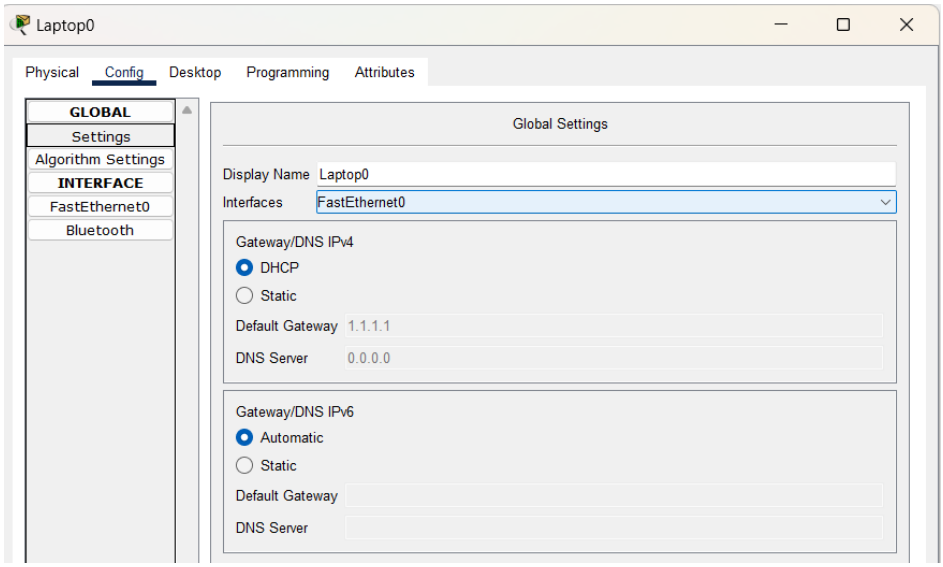


4. Wireless setting in Home Gateway:

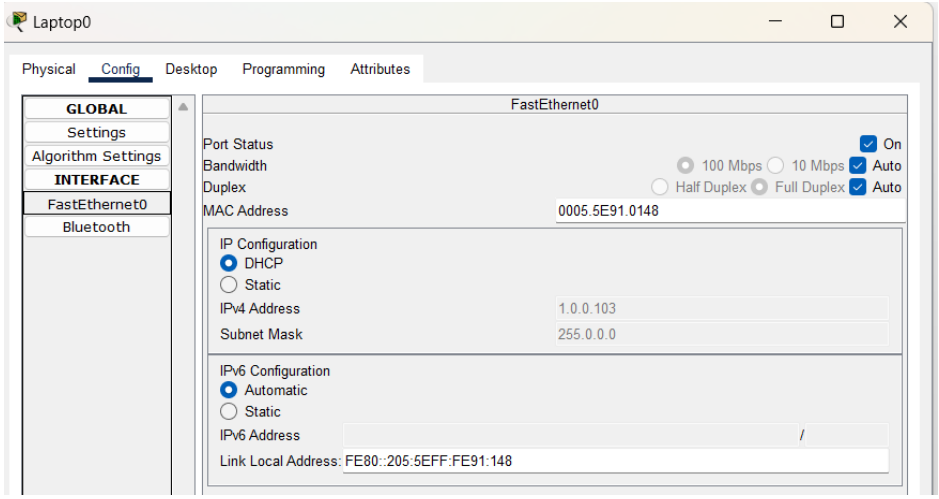


Step 5: Connect the laptop to the Home Gateway using a copper wire and configure the laptop:

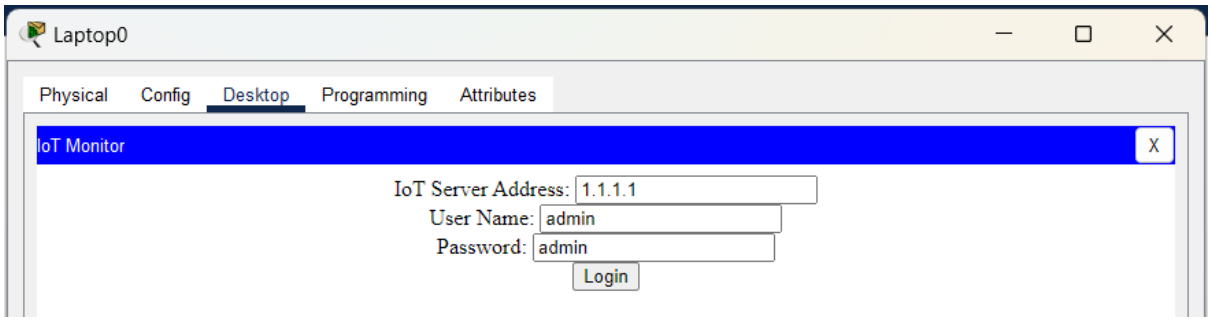
1. Global settings in Laptop:



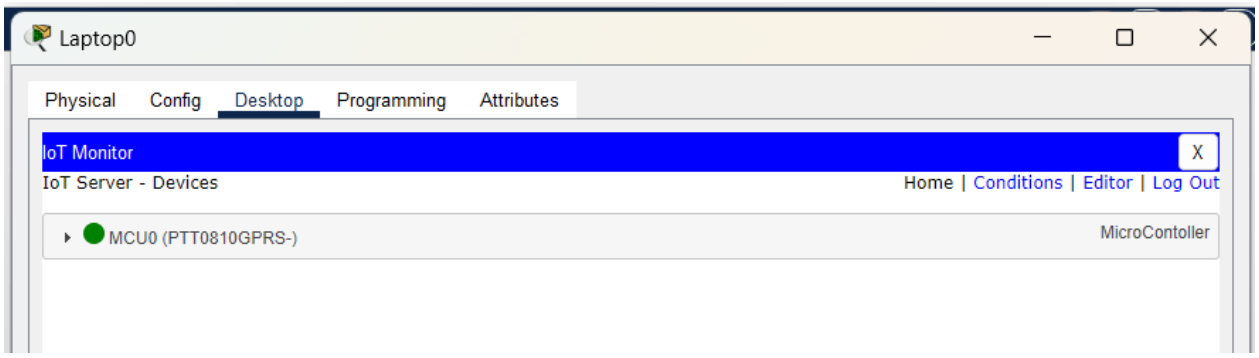
2. Fast Ethernet setting in laptop:



3. After configuring the laptop: Go to Desktop, open the IoT monitor and login via the Home Gateway Address.

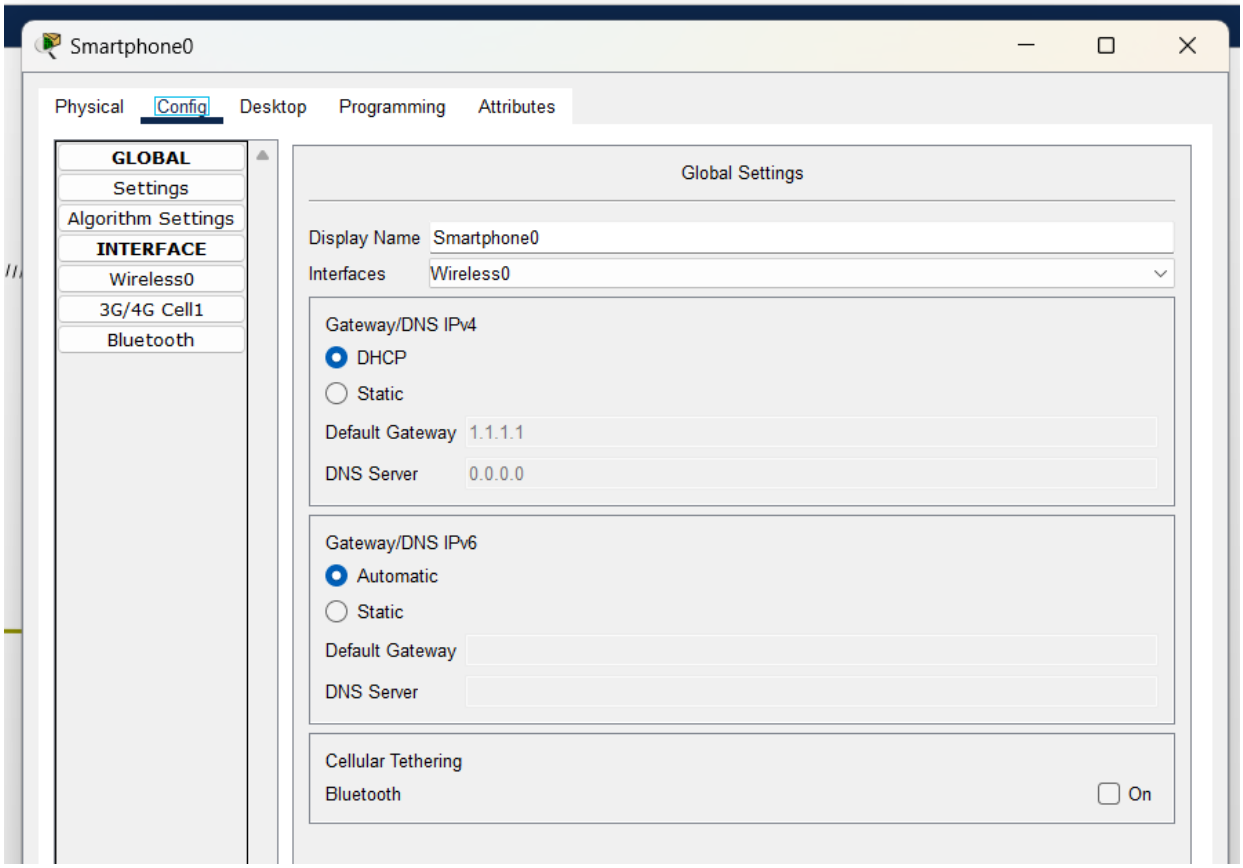


You will see the MCU in IoT Monitor

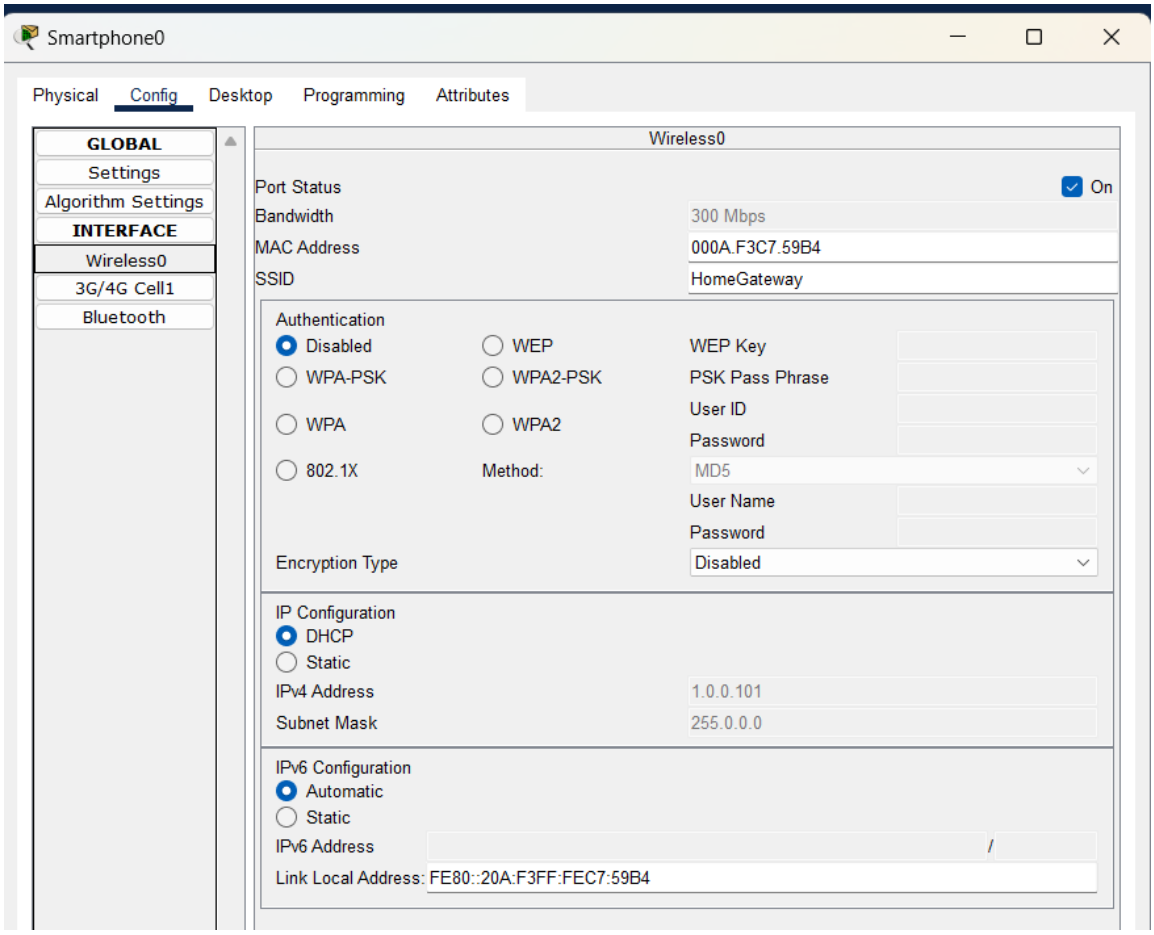


Step 6: Configure the smartphone:

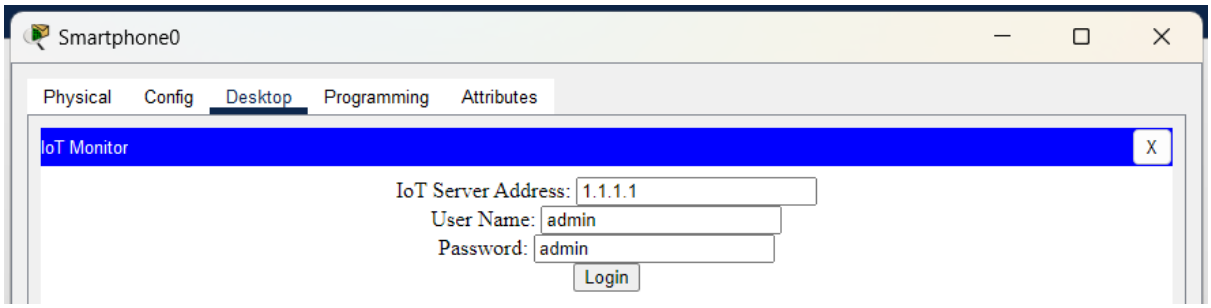
1. Global settings:



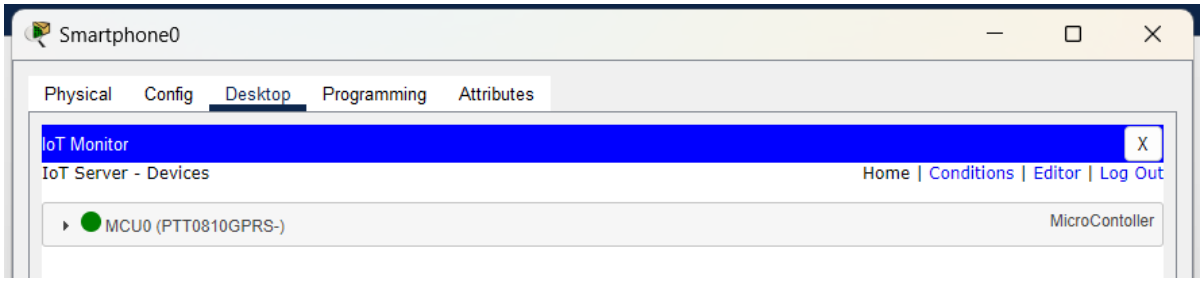
2. Wireless settings:



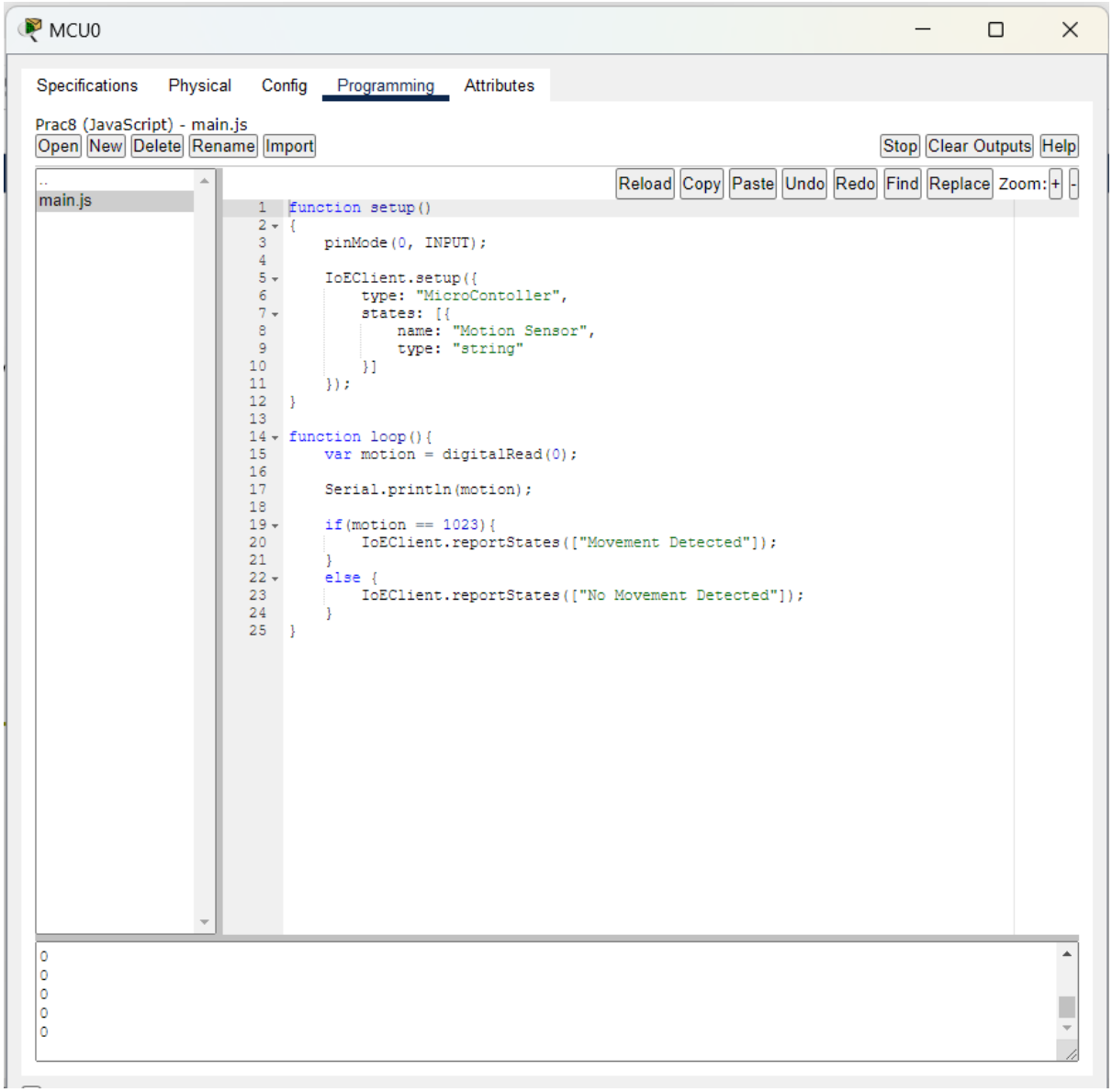
3. After configuring the smartphone: Go to Desktop, open the IoT monitor and login via the Home Gateway Address.

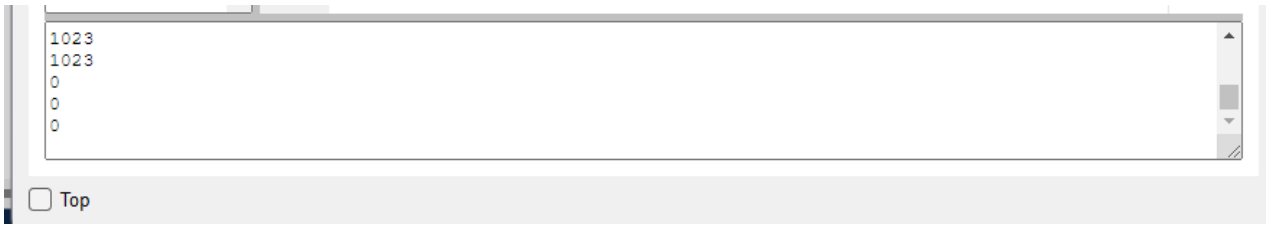


You will see the MCU in IoT Monitor



Step 7: Run the code which we wrote in MCU





We have to press alt key and hover the mouse over motion sensor. It will detect the motion.

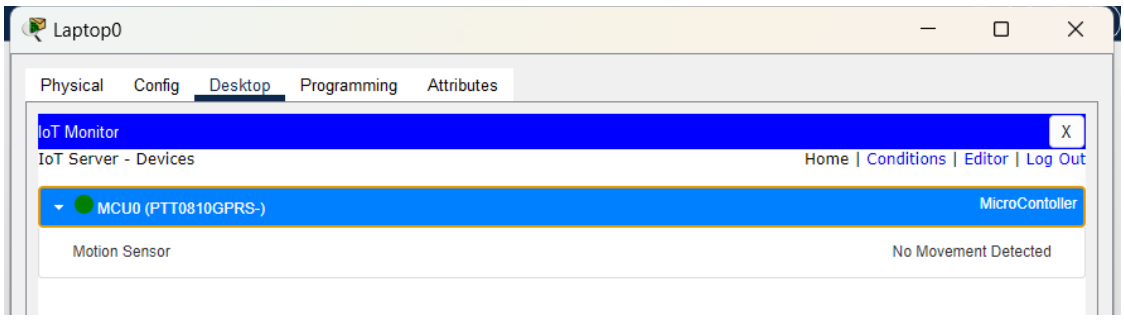
On the console of the Microcontroller, when there is no movement, the output shown is ‘0’ and when there is movement the output shown is ‘1023’.

Output:

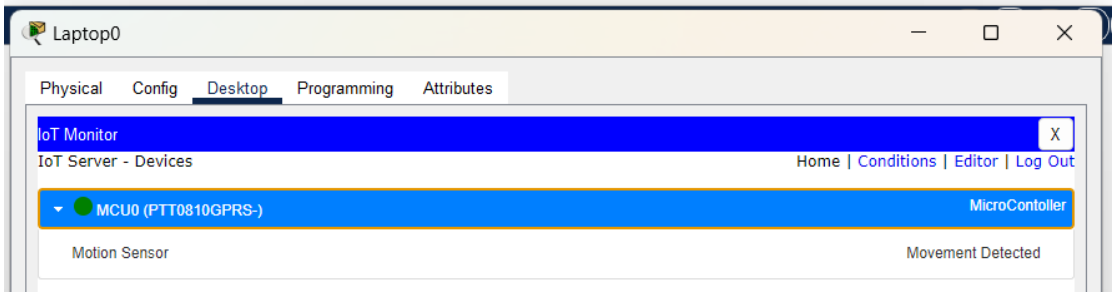
Check the movement in the IoT monitor of laptop and smartphone.

Laptop:

When no movement is detected.

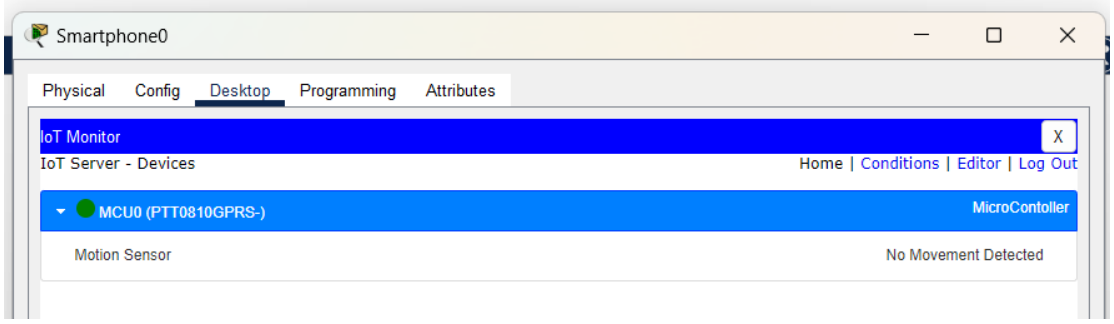
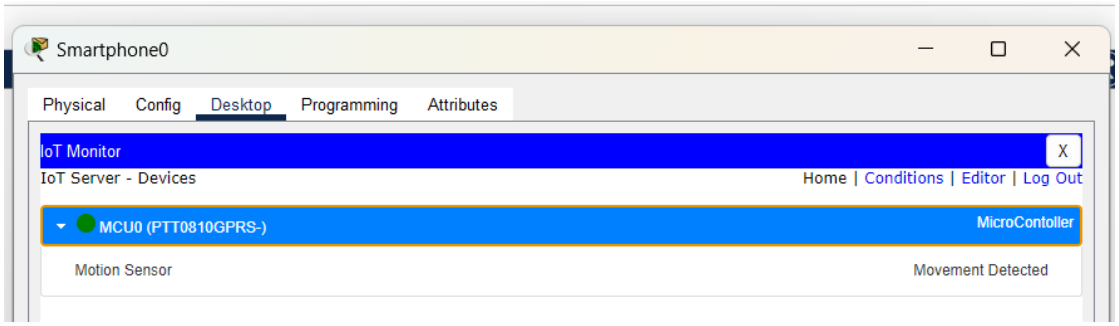


When movement is detected.



Smartphone:

When no movement is detected.



When movement is detected.